BetterCode: Readable

Write code you will understand six months later.

Why should ~~code~~ scientific articles be readable?

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Impacts of soret and dufour possessions on micropolar fluid past a stretching sheet in a porous medium

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Abstract

This work look for Soret and Dufour thing on MHD flow for micropolar fluid on sheet with hole stretch, also see hot from electric and warm from move, which old study no much talk. These hot and mix change speed, hot, spin, and mix of fluid in layer. The big math changes to small math by same change and solve in MATLAB BVP4C. Picture line and table show number for skin friction, Nusselt, Sherwood. We see more M make slow speed, light heat make thick hot layer, more micro make more spin near wall. Also, hot from electric and warm from move make more hot, so need in work like earth heat take, oil get, and hot change machine in hole thing.

Why should code be readable?

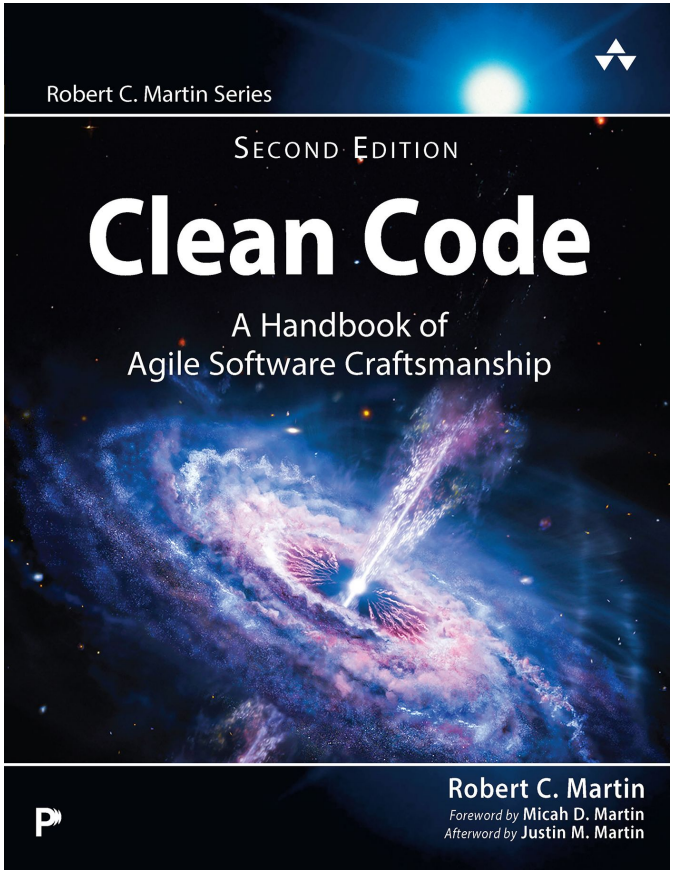
- Facilitate comprehension
- Efficient debugging
- Those who code clearly, speak clearly

What makes code readable?

- Clean Structure
- Meaningful Names
- Good Comments
- Neat Format

Good reference: Clean Code by Rober Martin

(Chapter 1, 2, 4, 5)



Demo 1: Folder Structure

https://github.com/ptuan5/better-code/blob/main/modules/day-01-readable/demos/folder-structure.md

Demo 1: Folder Structure

Data type	Workflow stage
study/ |-- README.md |-- data/ | |-- raw/ | `-- processed/ |-- scripts/ |  |-- data_process/ |  `-- pipeline/ |-- notebooks/ |-- results/ | |-- tables | |-- figures `-- logs	study/ |-- README.md |-- raw_data/ |-- preprocess/ | |-- cleaned_data | |-- scripts | `-- qc_reports |-- analysis-1/ |  |-- results |  `-- scripts |-- analysis-2 `-- …

Activity 1 - Organize a directory

https://github.com/ptuan5/better-code/blob/main/modules/day-01-readable/activities/activity-1/prompt.md

Activity 1 - Discussion

What structure choices do you prefer?

How would you name the files?

When is it worth creating more folders?

Demo 2: Readable code

https://github.com/ptuan5/better-code/blob/main/modules/day-01-readable/demos/readable-code.md

What does this code do?

```
take_res <- function(df, x_var) {
  res <- matrix(nrow = 0, ncol = ncol(df))
  colnames(res) <- colnames(df)
  for (i in 1:nrow(df)) {
    model <- lm(df[i,] ~ x_var)
    res <- rbind(res, model$residuals)
  }
  rownames(res) <- rownames(df)
  return(res)
}
```

Demo 2: Readable code

Some good tips:

- Meaningful name
- In-place and high-level comments
  - Roxygen for R and docstring for Python
- Linting (with Ruff)

Activity 2: Document a code chunk

https://github.com/ptuan5/better-code/blob/main/modules/day-01-readable/demos/readable-code.md

Activity 2 - Discussion

Which changes help the most?

Are the comments unnecessary?

What would you change if this script grows longer?

Take-away Message

- Better code helps others understand, trust, and adopt your research
- Readability helps with clear communication: clean structure & meaningful comments